

RISHABH SABHARWAL

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EDUCATION

MSc in Artificial Intelligence, University of Edinburgh, UK (August, 2026) | **Grade: Distinction***

B.Tech in ECE (AI & ML), NSUT, India (2024) | **GPA: 3.68/4** (CVSPK Fellow, **Top 5%**)

SELECTED PUBLICATIONS

Rishabh Sabharwal, H. Wang, A. Storkey, J. Pan, “Multi-Turn Evaluation of Deep Research Agents Under Process-level Feedback,” in SCALE Workshop, ICML 2026. (**Oral - Top 6%**) ([Link](#))

A.Mazumder, **Rishabh Sabharwal**, ..., P. Rathore, “On Convergence of Adam with Data Dependent Stepsize” in **IEEE Transactions on AI** (Q1 Journal, Impact Factor: 4.8). ([Link](#))

K. Annuranjana, N. Shrivastava*, A. Johar*, **Rishabh Sabharwal***, ..., “Does Reasoning Kill the Joke? Long-Context Humor Understanding in Hindi” in C3NLP Workshop, ACL 2026.

SELECTED PROJECTS

Multi-Turn Evaluation of Deep Research Agents (DRAs) ([Code](#))

- Designed a multi-turn evaluation framework and evaluated 3 LangChain-scaffolded DRAs using LLM-as-judge.
- Implemented observability and tracing via LangSmith to monitor tool usage, costs, and latency across runs.
- Exposed a reliability flaw in current deep research architectures:** the full-rewrite paradigm causes models to regress on up to **24%** of previously satisfied criteria. Models that implicitly preserve content incur **58%** higher costs and **50%** higher latency, making multi-turn deep research unreliable.

Multi-Step Agentic RAG System for Multi-Hop QA ([Code](#))

- Built a RAG pipeline for multi-hop QA over **270K** facts using FAISS, HuggingFace embedding models, and reranking; deployed via FastAPI with vLLM-served Qwen3-8B and GPT-4.1-mini for inference.
- Designed a decoupled **agentic RAG system** with multi-step retrieval via LLM tool calling, improving EM from **0.56** (single-step RAG) to **0.67**. Conducted systematic evaluation and error analysis using LLM-as-judge.

LLM Post-Training on Small-Scale Models ([Code](#))

- Fine-tuned Qwen2.5-0.5B on GSM8K via **SFT** and **RLHF**; mitigated reward over-optimization using process reward models assigning dense rewards to CoTs, **improving accuracy from 31% to 37%** on test set
- Designed a **latent-space knowledge distillation** framework transferring K latent reasoning tokens from Qwen2.5-7B directly into the student’s representation space, achieving **41% accuracy** on the test set.

EXPERIENCE

Research Engineer

Jan 2024 - Jul 2025

Indian Institute of Science

Bengaluru, India

- Developed ReGA, a parameter-free spectral GNN module for global context filtering. Achieved SOTA on 20 real-world datasets, reducing parameters by **55x** and GFLOPs by **100x**.
- Led a team to design a plug-and-play graph-diffusion-inspired attention module for ConvNets. Our module **adds just 20 parameters to ResNet-50 yet outperforms ResNet-101** (~20M more parameters); generalizes across ConvNeXt-V2 and MogaNet backbones.

SKILLS

- Programming & Frameworks:** Python, Pandas, NumPy, PyTorch, FastAPI
- ML/LLM Tooling:** Hugging Face, PEFT (LoRA, QLoRA), vLLM, Weights & Biases, GCP (Vertex AI)
- Agentic AI:** LangChain, LangSmith, Multi-Agent Systems, Tool Calling
- DevOps/Systems:** Git, Linux, CI/CD, Distributed Training (PyTorch DDP)